

Radius, Border, and Circle Area



Grade 7 • Mathematics
NC.7.G.4

Distinguish circumference and circle area.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A circular sign has radius 5 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

$$d = 2(5) = 10. \quad C = 2\pi(5) = 10\pi \approx 31.40 \text{ cm}. \quad A = \pi(5)^2 = 25\pi \approx 78.50 \text{ cm}^2.$$



Build a second model together

A circular sign has radius 6 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

Discuss: What should the first step find?



Your turn • 1

A circular sign has radius 9 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

Represent → solve → explain → check



Your turn • 2

A circular sign has radius 8 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

Represent → solve → explain → check



Your turn • 3

A circular sign has radius 5 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

Represent → solve → explain → check



Your turn • 4

A circular sign has radius 8 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

Represent → solve → explain → check



Exit question • Explain independently

A circular sign has radius 6 cm. Find its diameter, circumference, and area. Give exact answers using π , then approximate with $\pi = 3.14$.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 18 cm; 18π cm; 81π cm²

2. 16 cm; 16π cm; 64π cm²

3. 10 cm; 10π cm; 25π cm²

4. 16 cm; 16π cm; 64π cm²

Exit: 12 cm; 12π cm; 36π cm²

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

